

LAB-6

CCS 3104-G4

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code

- insert data

```
package com.example.lab6;

import java.sql.*;

public class InitDB {
    public static void main(String[] args) {
        String url =
            "jdbc:derby:javabook;create=true;user=scott;password=tiger";

        try (Connection conn = DriverManager.getConnection(url);
            Statement stmt = conn.createStatement()) {

            System.out.println("Database created/connected...");

            // 尝试删除旧表, 避免重复创建报错
            try { stmt.execute("DROP TABLE Scores"); } catch (SQLException
e) {}

            // 创建表
            stmt.execute("CREATE TABLE Scores (name varchar(20), score
double, permission boolean)");
            System.out.println("Table created...");

            // 插入数据
            stmt.execute("INSERT INTO Scores values ('John', 90.5, true)");
            stmt.execute("INSERT INTO Scores values ('Michael', 100,
true)");
            stmt.execute("INSERT INTO Scores values ('Michelle', 100,
false)");

            System.out.println("Data inserted successfully.");
        }
    }
}
```

```

        } catch (SQLException e) {
            e.printStackTrace();
        }
    }
}

```

- service interface

```

package com.example.lab6;

import java.rmi.Remote;
import java.rmi.RemoteException;

public interface ScoreService extends Remote {
    // 定义一个方法：输入名字，返回成绩
    double getScore(String name) throws RemoteException;
}

```

- serve

```

package com.example.lab6;

import java.rmi.Naming;
import java.rmi.RemoteException;
import java.rmi.registry.LocateRegistry;
import java.rmi.server.UnicastRemoteObject;
import java.sql.*;

public class ScoreServer extends UnicastRemoteObject implements ScoreService
{
    // 数据库连接配置
    private static final String DB_URL =
        "jdbc:derby:javabook;user=scott;password=tiger";

    protected ScoreServer() throws RemoteException {
        super();
    }

    @Override

```

```

public double getScore(String name) throws RemoteException {
    double score = -1.0; // 默认值, 表示未找到或无权限
    Connection conn = null;
    PreparedStatement pstmt = null;
    ResultSet rs = null;

    try {
        conn = DriverManager.getConnection(DB_URL);

        // 使用 PreparedStatement
        String sql = "SELECT score, permission FROM Scores WHERE name =
?";

        pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, name);

        rs = pstmt.executeQuery();

        if (rs.next()) {
            boolean hasPermission = rs.getBoolean("permission");
            if (hasPermission) {
                score = rs.getDouble("score");
            } else {
                System.out.println("Found student but permission is
false.");
                score = -2.0; // 没有权限查看
            }
        }

        } catch (SQLException e) {
            e.printStackTrace();
        } finally {
            // 关闭资源
            try { if (rs != null) rs.close(); } catch (Exception e) {}
            try { if (pstmt != null) pstmt.close(); } catch (Exception e) {}
            try { if (conn != null) conn.close(); } catch (Exception e) {}
        }
        return score;
    }

    public static void main(String[] args) {
        try {
            // 启动 RMI 注册表
            LocateRegistry.createRegistry(1099);

            ScoreServer server = new ScoreServer();
            Naming.rebind("ScoreService", server);

            System.out.println("Server is running and waiting for

```

```

queries...");
    } catch (Exception e) {
        e.printStackTrace();
    }
}
}

```

- client

```

package com.example.lab6;

import javafx.application.Application;
import javafx.geometry.Insets;
import javafx.geometry.Pos;
import javafx.scene.Scene;
import javafx.scene.control.Button;
import javafx.scene.control.Label;
import javafx.scene.control.TextField;
import javafx.scene.layout.GridPane;
import javafx.stage.Stage;

import java.rmi.Naming;

public class ScoreClient extends Application {

    private ScoreService scoreService;

    @Override
    public void start(Stage primaryStage) {
        // --- UI 布局设置 ---
        GridPane grid = new GridPane();
        grid.setAlignment(Pos.CENTER);
        grid.setHgap(10);
        grid.setVgap(10);
        grid.setPadding(new Insets(25, 25, 25, 25));

        Label nameLabel = new Label("Student Name:");
        TextField nameField = new TextField();
        Button btnGetScore = new Button("Get Score");
        Label resultLabel = new Label("Score will appear here");

        grid.add(nameLabel, 0, 0);
        grid.add(nameField, 1, 0);
    }
}

```

```

grid.add(btnGetScore, 1, 1);
grid.add(resultLabel, 1, 2);

// --- 按钮点击事件 ---
btnGetScore.setOnAction(e -> {
    String name = nameField.getText();
    try {
        // 连接 RMI 服务器
        if (scoreService == null) {
            scoreService = (ScoreService)
Naming.lookup("rmi://localhost/ScoreService");
        }

        // 调用远程方法
        double score = scoreService.getScore(name);

        // 处理结果
        if (score == -1.0) {
            resultLabel.setText("Student not found.");
        } else if (score == -2.0) {
            resultLabel.setText("No permission to view score.");
        } else {
            resultLabel.setText("Score: " + score);
        }

    } catch (Exception ex) {
        resultLabel.setText("Error: " + ex.getMessage());
        ex.printStackTrace();
    }
});

Scene scene = new Scene(grid, 350, 200);
primaryStage.setTitle("Student Score Client");
primaryStage.setScene(scene);
primaryStage.show();
}

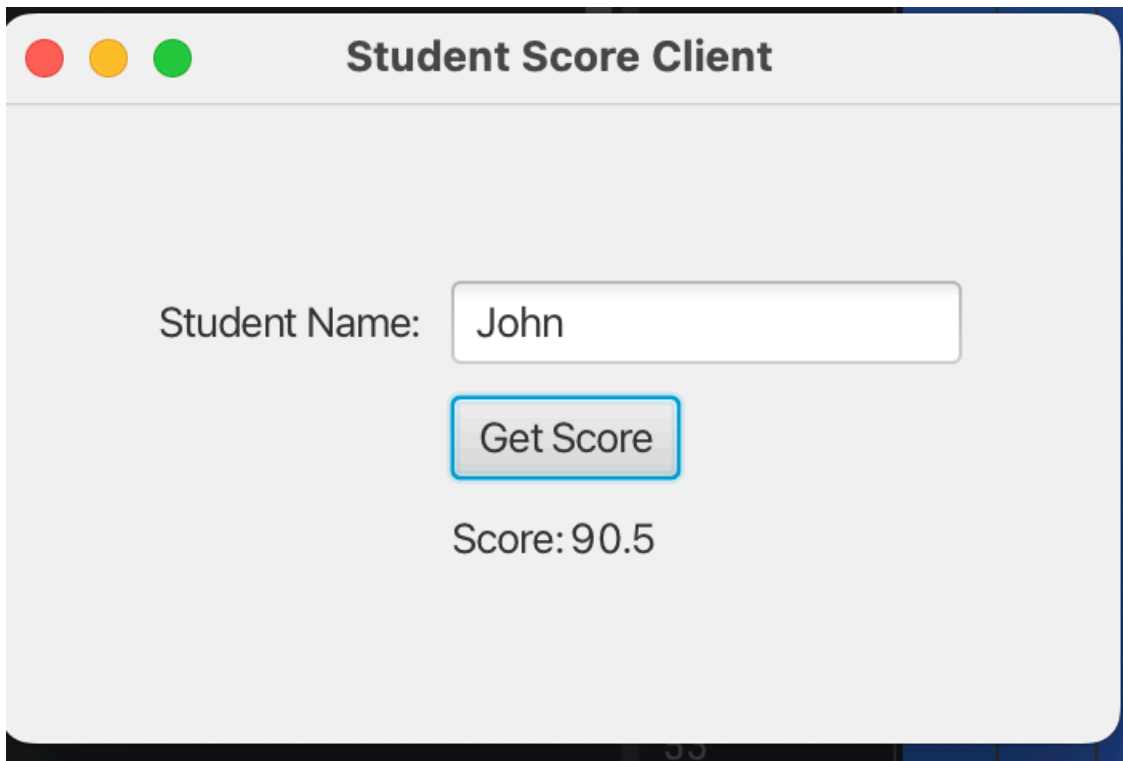
public static void main(String[] args) {
    launch(args);
}
}

```

- output

Student Name:

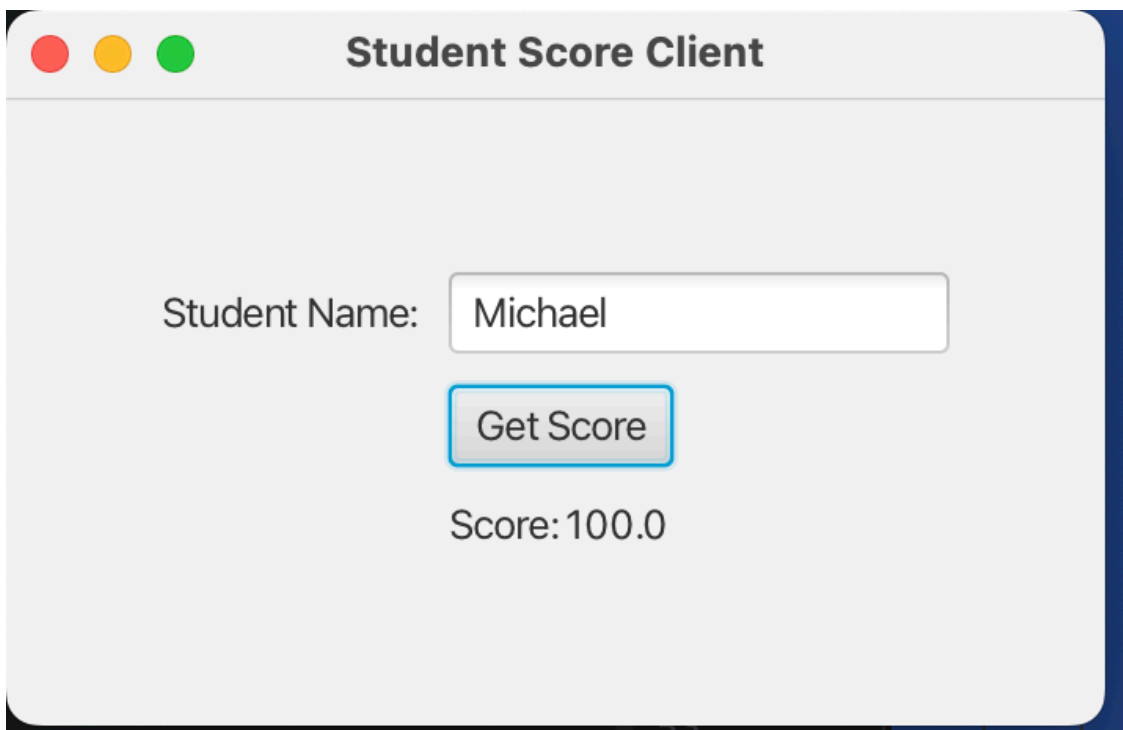
No permission to view score.



A screenshot of a macOS-style window titled "Student Score Client". The window has a light gray background and a title bar with three colored buttons (red, yellow, green) on the left. Inside the window, the text "Student Name:" is followed by a text input field containing "John". Below the input field is a button labeled "Get Score" with a blue border. Underneath the button, the text "Score: 90.5" is displayed.

Student Name:

Score: 90.5



A screenshot of a macOS-style window titled "Student Score Client". The window has a light gray background and a title bar with three colored buttons (red, yellow, green) on the left. Inside the window, the text "Student Name:" is followed by a text input field containing "Michael". Below the input field is a button labeled "Get Score" with a blue border. Underneath the button, the text "Score: 100.0" is displayed.

Student Name:

Score: 100.0